VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE

Matunga, Mumbai-400 019

Autonomous Institute affiliated to University of Mumbai

EXAMINATION	Mid Semester Examination March 2024	DATE OF EXAM	11/03/2024
SEMESTER & PROGRAM	Sem-II, First Year MCA	TIME	11.15 am – 12.45pm
TIME ALLOWED	**1 1/2 HR**	MARKS	**40**
COURSE NAME – (CODE)	Cloud Computing – (R5MC5113S)		

Instructions

1. All questions carry equal marks.
2. Figures to the right indicate full marks.
3. Assume suitable data if required.

Q.1	a.	Define cloud computing? Elaborate on the importance of distributed computing, offering valid justifications.	(5)
	b.	Which are various 5 essential attributes of Cloud Computing? Discuss in detail.	(5)
	c.	What is Grid computing? Detail various applications where the distributed paradigm of Grid computing finds relevance.	(5)
	d.	Differentiate between HTML and XML. Give advantages of XML.	(5)
Q.2	a.	Which are the different 7 steps for migration to cloud? Explain in detail.	(7)
	b.	What is a virtualization? Explain OS assisted (paravirtualization) in detail with diagram.	(5)
	c.	What are the different IaaS characteristics? Which are the various applications where IaaS may be useful?	(5)
	d.	Which are the different scenarios where SaaS may be useful?	(3)
		All the best	

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EXAMINATION	**Mid Sem Exam March 2024**	DATE OF EXAM	**13/03/2024**
SEMESTER & PROGRAM	**Sem - II, First Year MCA**	TIME	**11.15 am to 12.45 pm**
TIME ALLOWED	**1&1/2 HRS.**	MARKS	**40**
COURSE NAME – (CODE)	**DESIGN AND ANALYSIS OF ALGORITHM – (R5MC5022T)**		

Instructions
1. All questions carry equal marks.
2. Figures to the right indicate full marks.
3. Assume suitable data if required.

Q.1 a. Which sorting algorithm is best if the list is already sorted? Why? (2)

b. Write the time complexity for each line of the following algorithm. (Do not consider each line as separate line of code) (4)

```
void function(int n)
{
    int count = 0;
    for (int i=n/2; i<=n; i++)
        for (int j=1; j<=n; j = 2 * j)
            for (int k=1; k<=n; k = k * 2)
                count++;
}
```

c. What is the divide and conquer technique? Explain in detail. (4)

Q.2 a. What is stable sorting and give example of stable and unstable sorting algorithms. (4)

b. Write the algorithm for Quick sort. (6)

Q.3 a. A machine needs on average 200 sec to sort 1000 elements by Quick sort. Calculate the minimum time needed to sort 200 elements. (4)

b. What are the characteristics of an algorithm? Explain in detail. (6)

Q.4 a. Sort the following elements using insertion sort algorithm. (10)

12, 29, 25, 8, 32, 17, 40

* Best of Luck *

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[Central Technological Institute, Maharashtra State]
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EXAMINATION	Mid Semester Examination March 2024	DATE OF EXAM	Monday 18.03.2024
SEMESTER & PROGRAM	2nd & MCA	TIME	11.15 to 12.45
TIME ALLOWED	1.5 Hrs	MARKS	40
COURSE (Course Code)	R5MC5123T Software Design and Pattern		

Instructions: All questions are Compulsory

Q. No	Sub-ques	Question	Marks	CO
Q.1	a	With pattern description parameters explain The Abstraction-Occurrence Pattern	5	CO1
	b	Why Software Engineers Must Develop Understanding Of Users?	5	CO1
Q.2	a	Differentiate between Usability and Utility with reference to software design pattern.	5	CO2
	b	Consider software is to be built for automatic recording of online meeting attendance, assume suitable data and perform use case modelling.	5	CO2
Q.3	a	Online teaching software is to be built which should have following major functionalities, 1. User interface and control 2. Creating meeting, inviting participants 3. Presenting screen along with audio and video 4. Recording and storing in database 5. Computer graphics along with display 6. Taking attendance 7. Giving quiz and assignment 8. Peripherals attachment / external hardware attachment Perform ER modelling, assume suitable data.	5	CO3
	b	What are context, problem, forces, and solution of singleton pattern?	5	CO3
Q.4	a	What are Difficulties and Risks in Use Case Modelling and User Interface Design?	5	CO2
	b	What are Principles Leading to Good Design (list them)? Explain Increasing cohesion.	5	CO1

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EXAMINATION	Mid Semester Examination March 2024	DATE OF EXAM	18 /03/2023 Monday
SEMESTER & PROGRAM	Sem - II, Second Year MCA	TIME	11.15 am to 12.45 pm
TIME ALLOWED	1 & 1/2 HRS.	MARKS	40
COURSE NAME – (CODE)	Mathematical and statistical foundation 2– (R5MC5023S)		

Instructions
1. All questions carry equal marks.
2. Figures to the right indicate full marks.
3. Assume suitable data if required.

Q1	a	What is FA finite automata write formal definition of finite automata Draw the finite automata model digram and explain input tape finite control	(3)	CO2
	b	Draw the transition digram(graph) and write Transition table represent the FA or DFA or NFA is M=(Q,E,d,q0,F) i.e M=({q0,q1,q2},{0,1},d,q0,{q2}) where d is transition function given by example 1) d(q0,0)=q1 d(q1,0)=q0 d(q2,0)=q2 d(q0,1)=q2 d(q1,1)=q2 d(q2,1)= q2 example 2) d(q0,0)=q1 d(q1,0)=q1 d(q2,0)=q2 d(q0,1)=q1 d(q1,1)=q2 d(q2,1)=q2	(3)	CO2
	c	What is abstract algebra. Explain group ring field and Write properties of group ring field	(4)	CO1
Q2	a	What is CFG context free grammar 1. given CFG and generate the string **0011** of given grammar CFG=(V,T,P,S) CFG=({S}{0,1},P,S) Production rule is 1 S=0S 2 S=1S 3 S= e epsilon	(5)	CO2
		2. given CFG and Generate the string **aaabbbbbb** of given grammer CFG=(V,T,P,S) CFG=({S}{a,b},P,S) Production rule: 1 s=aSbb 2 S=abb 3.given CFG and generate the string **aabbb** of given grammer CFG=(V,T,P,s)		

		CFG=({S} {a,b},P,S Production rule: 1 S= ASB 2 S= e epsilon 3 A=a 4 B=bb 5 B=b		
	b	What is derivation tree or parse tree 1 draw the derivation tree of string **bbabb** using following production rule Given production rule is 1. S=bSb 2. S=a 3. S=b 2.draw the derivation tree **aabbbb** using following production rule Given production rule is 1 S= AB 2 S= e epsilon 3 A=aB 4. B=Sb 3.draw the derivation tree of string **abb** using following production rule Given production rule is 1 S=AB 2 S=e epsilon 3 A=aB 4 B=Sb	(5)	CO2
Q3	a	Write difference between DFA deterministic finite automata and NFA nondeterministic Finite automata	(5)	CO2
	b	What is regular expression Write the 4 rule of regular expression 1)Kleen closure(*) 2)positive closure(+) 3)concatenation(.) 4)union(U) Draw the finite automata of the given regular expression is 1) Phi 2) e-epsilon 7) a* 8) b+ 3) a 4) b 9) 01 10) 0+1 5) a b 6) a + b	(5)	C02
Q4	a	1. What are types of machine learning & explain it. And write application of machine learning 2. Explain normal distribution and write formula and characteristics of it 3. What is GSM Gaussian mixture model draw diagram Explain with example and what are the advantages & disadvantages of GMM	(6)	C04
	b	What is SVM support vector machine how does work what are types of svm draw suitable digram What is hyperplane what is support vector What is margine	(4)	CO5

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EXAMINATION	Mid Semester Examination March 2024	DATE OF EXAM	Wednesday 20.03.2024
SEMESTER & PROGRAM	Sem-II, First Year MCA	TIME	11.15 am – 12.45pm
TIME ALLOWED	**1 1/2 HR**	MARKS	**40**
COURSE NAME – (CODE)	Data Mining – (R5MC5021T)		

Instructions
1. All questions carry equal marks.
2. Figures to the right indicate full marks.
3. Assume suitable data if required.

Q.1 a. What is K-NN classifier? Apply to the following new test data instance to classify it properly. (5)

S. N	Sepal Length	Sepal Width	Species
1	5.3	3.7	Setosa
2	5.1	3.8	Setosa
3	7.2	3.0	Virginica
4	5.4	3.4	Setosa
5	5.1	3.3	Setosa
6	5.4	3.9	Setosa
7	7.4	2.8	Virginica
8	6.1	2.8	Verscicolor
9	7.3	2.9	Virginica
10	6.0	2.7	Verscicolor
11	5.8	2.8	Virginica
12	6.3	2.3	Verscicolor
13	5.1	2.5	Verscicolor
14	6.3	2.5	Verscicolor
15	5.5	2.4	Verscicolor
Test Instance	5.2	3.1	?

b. What is Naive Bayes classifier? Apply to the following new test data instance to classify it properly. (5)

S.N.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes

6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes
Test Instance	Red	SUV	Domestic	?

Q.2 a What is data mining? Explain knowledge discovery in data process. (5)

b Consider the following table (5)

Customer ID	Transaction ID	Items Bought
1	1	{a,d,e}
1	2	{a,b,c,e}
2	3	{a,b,d,e}
2	4	{a,c,d,e}
3	5	{b,c,e}
3	6	{b,d,e}
4	7	{c,d}
4	8	{a,b,c}
5	9	{a,d,e}
5	10	{a,b,e}

Taking transaction ID as a market basket find support for each itemset {a}, {c,d}, and {b,d,e}

Q.3 a What is sampling, List and explain various sampling techniques. (5)

b Consider a table given below (5)

T ID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Find the confidence of

1. {Diaper} -> {Beer},
2. {Milk, Bread} -> {Eggs,Coke},
3. {Beer, Bread} -> {Milk}

Q.4 a What are different types of attributes? List and explain properties of attributes. (5)

b With the help of suitable example, explain the statement (5)

$\forall$ a,b: if $a \subseteq b \Rightarrow s(a) >= s(b)$, where 's' is the support